



AI-Based Cyber Attack Prediction & Defense System

SECURITY INCIDENT REPORT

Generated: 15 May 2026 07:38 UTC

■ CLASSIFICATION: CONFIDENTIAL

Total Flows Analysed	Attack Events
100	10
Benign Flows	Attack Rate
90	10.0%
IPs Blocked	Attack Types Found
1	2

1. EXECUTIVE SUMMARY

AEGIS processed **100** network flows during the reporting period. Of these, **10** were classified as malicious (10.0% attack rate). The hybrid detection engine (Random Forest ML + rule-based heuristics) identified **2** unique attack categories. The automated defense layer blocked **1** source IPs.

Total Flows	Total Attacks
100	10
Benign Traffic	Attack Rate
90	10.0%
Blocked IPs	Attack Types
1	2

2. ATTACK TIMELINE (1 EVENTS)

Timestamp	Source IP	Attack Type	Confidence	Action
05-15 07:00:00	1.2.3.4	PortScan	100.0%	BLOCKED

3. ATTACK TYPE DISTRIBUTION

PortScan		7 (70.0%)
DDoS		3 (30.0%)

4. DEFENSE ACTIONS — BLOCKED IPS

IP Address	Status
1.2.3.4	BLOCKED

5. TOP RISK IPS

IP Address	Risk Score	Block Count	Country	Last Seen
1.2.3.4	91	3	CN	2026-05-15T07:00

6. MODEL PERFORMANCE

The AEGIS ML engine uses a Random Forest classifier trained on the CICIDS2017 dataset (2,226,224 training samples, 18 flow features). Performance metrics below reflect evaluation on the held-out test set (445,245 samples).

Accuracy	Precision
98.34%	98.12%
Recall	F1 Score
97.98%	98.05%

Per-Class Metrics

Attack Class	Precision	Recall	F1
BENIGN	100%	97%	99%
DDoS	100%	100%	100%
DoS GoldenEye	88%	100%	94%
DoS Hulk	97%	100%	99%
DoS Slowhttptest	99%	99%	99%
DoS slowloris	100%	99%	100%
FTP-Patator	100%	100%	100%
Heartbleed	100%	100%	100%
PortScan	100%	100%	100%
SSH-Patator	87%	100%	93%
Web Attack Brute Force	17%	57%	26%
Web Attack Sql Injection	33%	25%	29%
Web Attack XSS	6%	68%	10%

7. RECOMMENDATIONS

- Deploy rate-limiting at network edge (BPS cap per source IP). Consider upstream provider DDoS scrubbing (e.g., Cloudflare Magic Transit).
- Enable stateful packet inspection on firewall. Silently drop RST/SYN packets on closed ports. Use PSAD for real-time PortScan detection.
- Block rate is below 80%. Review geo-block lists and expand automated blocking policy to cover HIGH-confidence ML detections.

This report was automatically generated by the AEGIS Cyber Defense System. All detections are based on the CICIDS2017-trained Random Forest model combined with rule-based heuristics. Treat this document as CONFIDENTIAL.